STUDENT PERFORMANCE & ACADEMIC ANALYTICS

Exploratory Data Analysis (EDA) Using Python

PROJECT REPORT

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Dataset: UCI Student Performance Dataset

Project Domain: Data Analytics

Tools & Technologies:

- Python
- Pandas
- Matplotlib
- Seabor
- Jupyter Notebook

Project Objective

To explore student academic performance and identify patterns and relationships between final grades and factors such as study time, previous failures, family relationships, and internet access.

Academic Project Report — 2026

1. INTRODUCTION

Exploratory Data Analysis (EDA) is an important process in data analytics. It helps us understand a dataset by examining its structure, identifying patterns, finding relationships between variables, and discovering useful insights through statistical methods and visualizations.

The project titled "Student Performance & Academic Analytics" focuses on analysing students' academic performance using Python. The analysis is based on the UCI Student Performance Dataset, which contains information about students' demographic backgrounds, family conditions, study habits, and academic results.

The final grade (G3) is considered the target variable for understanding academic performance. Different factors, including study time, previous failures, family relationships, and internet access, are explored to identify possible patterns associated with students' final grades.

2. PROJECT OBJECTIVES

The main objectives of this project are:

1. To understand and explore the Student Performance Dataset.

2. To analyse the distribution of students' final grades (G3).

3. To study the relationship between study time and academic performance.

4. To examine the impact of previous academic failures on final grades.

5. To analyse the relationship between family conditions and student performance.

6. To compare academic performance based on internet access.

7. To identify patterns, trends, and outliers using data visualization.

8. To summarize meaningful insights from the exploratory analysis.

3. SCOPE OF THE PROJECT

This project focuses on exploratory analysis rather than predicting student grades. It examines selected academic and social factors to understand how they are associated with students' final academic performance.

4. DATASET DESCRIPTION

The dataset used in this project is the UCI Student Performance Dataset. It contains information about students' demographic characteristics, family background, study habits, social activities, and academic performance.

Dataset Overview:

- Dataset Name: Student Performance Dataset

- Source: UCI Machine Learning Repository

- Number of Records: 395

- Number of Features: 33

- Target Variable: G3 (Final Grade)

- File Format: CSV

Important Variables:

Variable| Description

school| Student's school

sex| Gender of the student

age| Age of the student

studytime| Weekly study time

failures| Number of past class failures

family| Family relationship quality

internet| Internet access at home

G1| First period grade

G2| Second period grade

G3| Final grade

5. DATA CLEANING AND PREPROCESSING

Data cleaning is an important step in Exploratory Data Analysis. It helps improve data quality and makes the dataset suitable for analysis.

The following steps were considered during preprocessing:

1. Loading the Dataset: The CSV dataset was imported using the Pandas library.

2. Understanding Dataset Structure: The shape, column names, data types, and general information were examined.

3. Checking Missing Values: The dataset was checked for missing or null values.

4. Checking Duplicate Records: Duplicate entries were checked to understand whether repeated records were present.

5. Statistical Summary: Descriptive statistics such as mean, median, standard deviation, minimum, and maximum values were examined.

6. Data Visualization: Graphs and plots were used to understand distributions and relationships between variables.

6. EXPLORATORY DATA ANALYSIS AND RESULTS

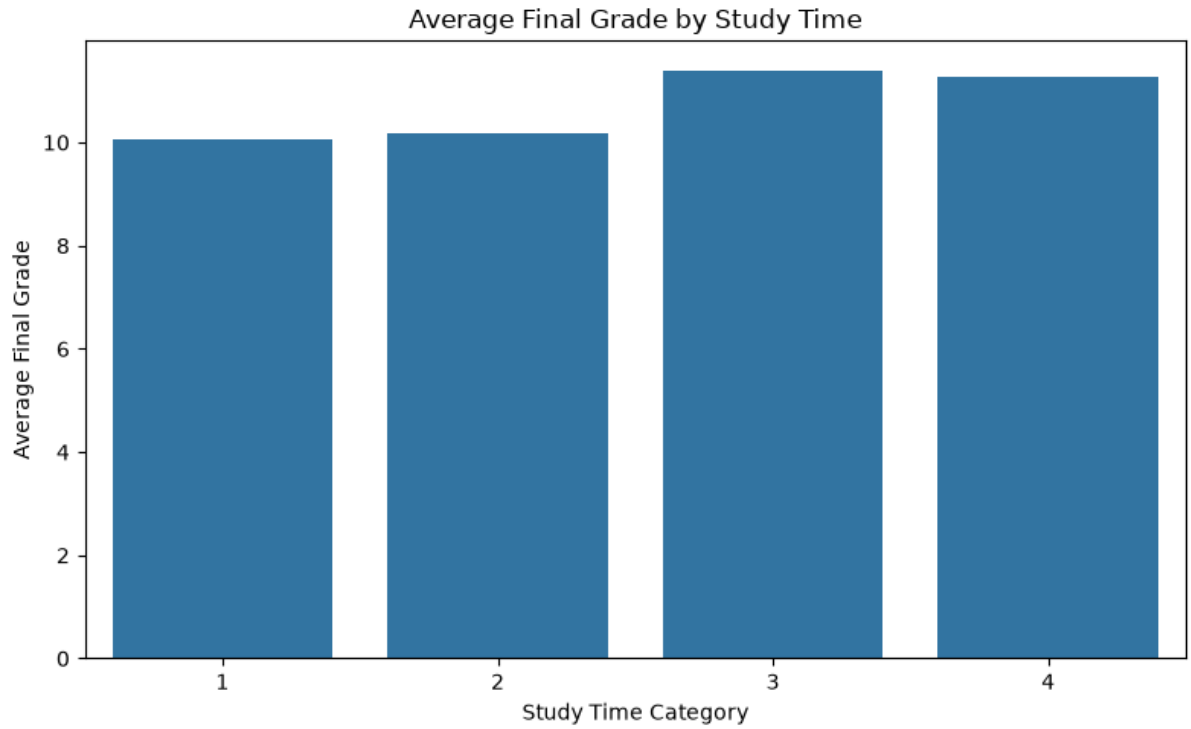
Exploratory Data Analysis was performed to understand the relationship between students' academic performance and different academic and social factors.

The final grade (G3) was used as the primary variable for comparing student performance across different groups.

6.1 Study Time and Academic Performance

Study time is an important factor that may be associated with academic performance. A boxplot was used to compare final grades (G3) across different weekly study-time categories.

The analysis helps examine differences in grade distributions, medians, variations among students with different study-time levels.

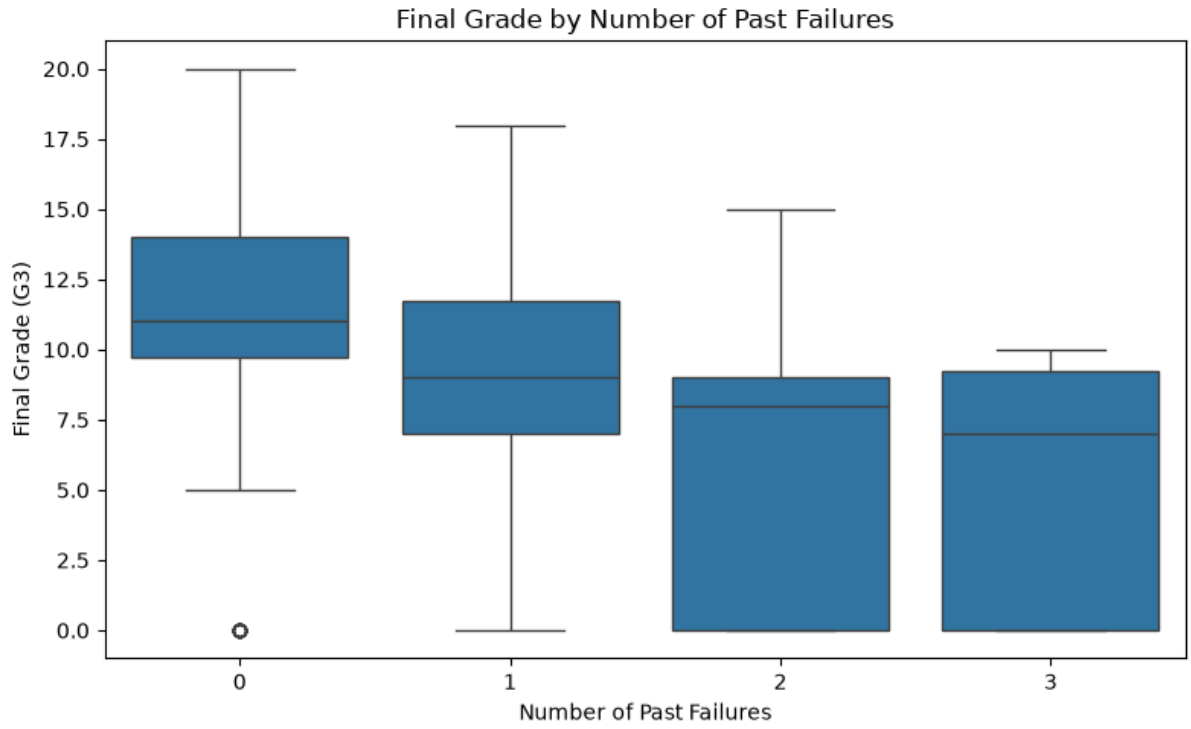


6.2 Previous Failures and Final Grades

Previous academic failures were examined to understand their association with students' final grades.

A boxplot was created using the number of past failures as the grouping variable and G3 as the target variable.

The visualization indicates that students with more previous failures generally tend to have lower final grades. This suggests that previous academic difficulties may be when useful identifying students who need additional academic support.



6.3 Family Relationship and Academic Performance

Family relationship quality was compared with students' final grades using group-wise statistical summaries and a boxplot.

The following table presents the calculated results:

Family Rating| Students| Mean G3

1| 8| 10.63

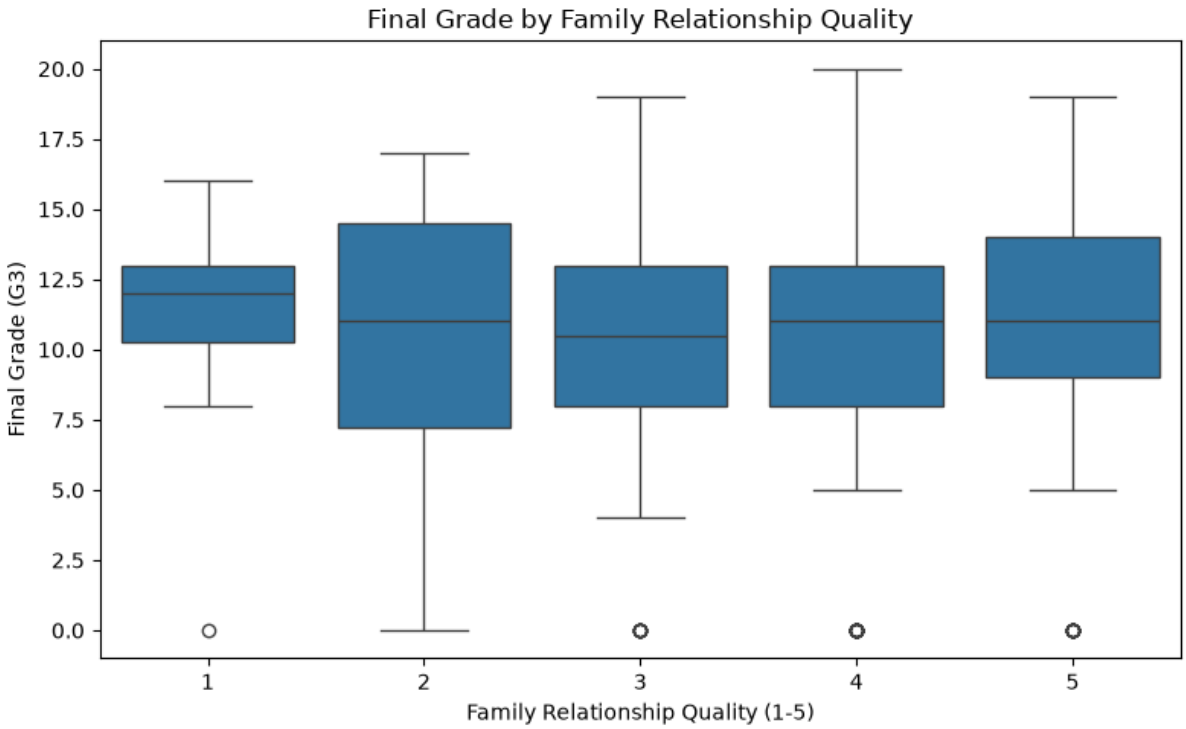
2| 18| 9.89

3| 68| 10.04

4| 195| 10.36

5| 106| 10.83

The average final grades show relatively modest differences across family relationship categories. The highest average in this summary is observed for rating 5.



6.4 Internet Access and Academic Performance

Internet access was analysed by comparing the average final grades of students with and without internet access at home.

The calculated results are:

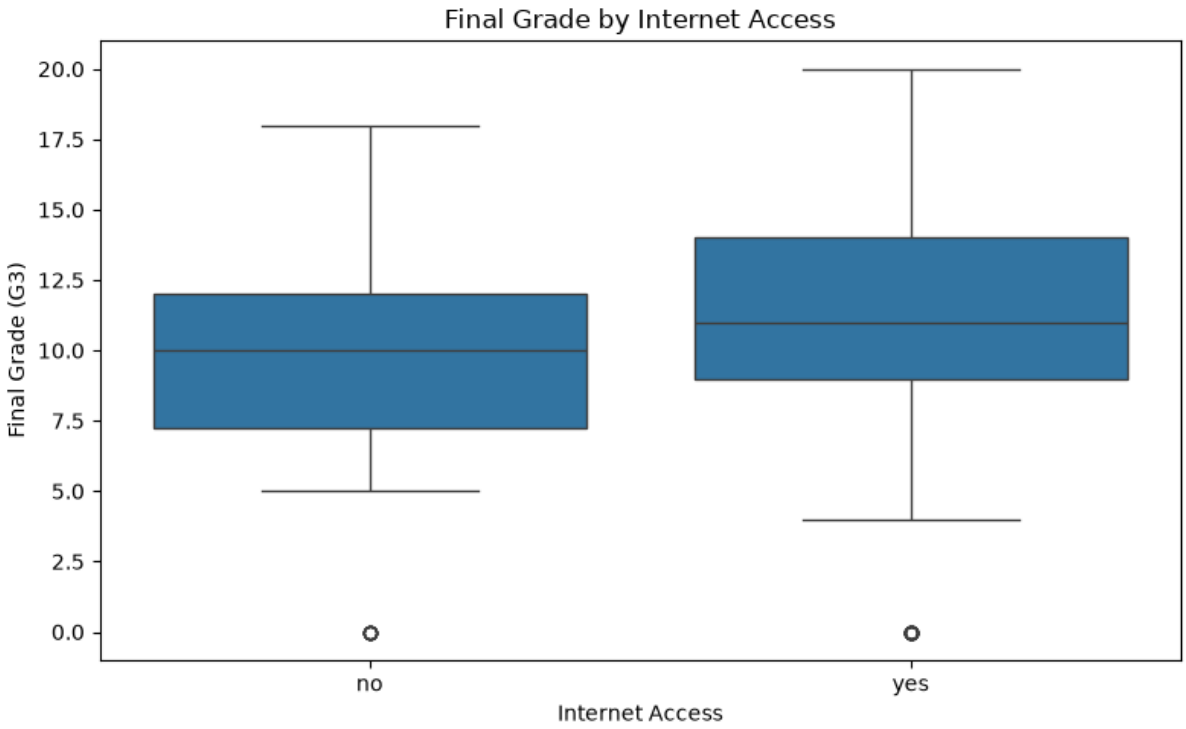
Internet Access| Number of Students| Mean G3

No| 66| 9.41

Yes| 329| 10.62

Students with internet access have a higher average final grade in this dataset. The difference between the two groups means is approximately 1.21 grade points.

Internet access may provide better access to educational resources, but this comparison does not establish that internet access directly causes higher grades.



7. KEY FINDINGS

The exploratory data analysis produced the following important observations:

1. Previous Academic Failures: Students with a higher number of previous failures generally showed lower final grades.

2. Study Time: Final grade distributions varied across weekly study-time categories. However, study time alone cannot explain academic performance.

3. Family Relationships: Students with different family relationship ratings showed relatively small differences in average final grades.

4. Internet Access: Students with internet access had an average final grade of 10.62, compared with 9.41 for students without internet access.

5. Dataset Overview: The analysis was performed on 395 student records with 33 features.

These findings describe patterns in the dataset and do not establish direct cause-and-effect relationships.

8. LIMITATIONS OF THE STUDY

1. The dataset contains information from a specific educational context and may not represent all students.

2. The analysis is based on observational data, so causal relationships cannot be established.

3. Some categories contain fewer students than others, which may affect group comparisons.

4. Academic performance may depend on several factors that are not fully captured by the dataset.

5. The findings are based on the available dataset and its recorded variables.

9. CONCLUSION

This project successfully explored student academic performance using Exploratory Data Analysis (EDA) with Python.

The UCI Student Performance Dataset was analysed using Pandas, Matplotlib, and Seaborn. Different statistical summaries and visualizations were used to examine the relationship between final grades and factors such as study time, previous academic failures, family relationships, and internet access.

The analysis highlighted useful patterns in student performance. In particular, previous academic failures showed a noticeable association with lower final grades, while students with internet access had a higher average final grade in the dataset.

This project improved practical understanding of data cleaning, statistical analysis, data visualization, and interpretation of real-world datasets.

Overall, EDA is a valuable approach for discovering patterns and generating insights that can support better understanding of academic performance.

10. REFERENCES

1. UCI Machine Learning Repository – Student Performance Dataset.
https://archive.ics.uci.edu/dataset/320/student+performance

2. Pandas Documentation.

https://pandas.pydata.org/docs/

3. Matplotlib Documentation.

https://matplotlib.org/stable/

4. Seaborn Documentation.

https://seaborn.pydata.org/

5. Python Documentation.

https://docs.python.org/3/